Precision Care Challenge 2026

Title: Cloud-Native AI-Ready Imaging Platform for Early Disease Triage

Problem Statement

Medical imaging is central to modern clinical decision-making, yet healthcare institutions face growing challenges in managing and utilizing imaging data effectively. Increasing imaging volumes, radiologist workload, fragmented on-premise infrastructure, and delayed identification of urgent cases collectively impact patient outcomes.

Cloud-native healthcare imaging services now offer the opportunity to redesign traditional workflows into scalable, secure, and AI-ready platforms. Such platforms can assist clinicians by prioritizing imaging studies that require timely attention, while ensuring data security, compliance, and ethical usage.

Challenge Objective

Design and implement a cloud-native medical imaging platform that enables ingestion, storage, retrieval, and AI-assisted triage of medical imaging data. The solution should leverage managed cloud services and demonstrate how imaging workflows can be modernized to support early care prioritization.

The solution is intended to function as a clinical decision-support system and must not be positioned as a diagnostic tool.

Expected Solution Capabilities

1. Medical Imaging Ingestion and Storage

- Ingest publicly available or simulated DICOM imaging datasets.
- Store and index imaging studies using AWS Healthcare Imaging.
- Enable efficient, low-latency access to imaging data through APIs.

2. AI-Assisted Triage Workflow

- Apply AI models or rule-based logic to flag and prioritize potentially urgent imaging studies.
- Provide confidence indicators and explainability for triage outputs.
- Use open-source or publicly available models and clearly indicate non-diagnostic usage.

3. Clinician-Facing Web Application

- Provide a secure web interface for clinicians to review imaging studies.
- Display DICOM images using an integrated medical image viewer.
- Surface triage indicators and allow sorting and filtering by priority.

4. Security and Compliance Readiness

- Implement role-based access controls.
- Ensure encryption in transit and at rest.
- Maintain audit logs for access and actions.
- Avoid the use of real patient data or proprietary datasets.

Constraints and Guidelines

- Only public, synthetic, or openly licensed datasets may be used.
- The architecture must be cloud-native and demonstrate scalability.
- Ethical considerations and clinical disclaimers must be clearly stated.

Reference Cloud Architecture

1. Imaging data is ingested from public datasets or simulated imaging sources.
2. An ingestion service uploads and manages imaging data in AWS Healthcare Imaging.
3. Stored imaging studies are indexed and made available for retrieval.
4. An AI-assisted triage layer processes studies to assign priority indicators.
5. A clinician-facing web application presents images, triage information, and audit logs securely.

Outcome

This challenge encourages participants to design cloud-native medical imaging platforms that reflect real-world healthcare constraints. The solutions should demonstrate technical depth, ethical awareness, and a clear path toward adoption in modern healthcare environments.